Fiscal Year:	FY 2005	**Task Last Updated:**	FY 11/01/2005
PI Name:	Sayler, Gary Ph.D.		
Project Title:	Prototype Biophotonic Biosensor for Monitoring VOC Contaminants in Spacecraft Habitats		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	37996	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	06/30/2004	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ripp, Steven (University of Tennessee) Islam, Syed (University of Tennessee) Blalock, Benjamin (University of Tennessee) Garland, Jay (Dynamac Corp.)		
Grant/Contract No.:	NNJ04HF02A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Bioluminescent Bioreporter Integrated Circuits (BBICs) have been developed for sensing a wide range of chemical and physical agents and it is the goal of this research to carry this to a prototype development of a generalized BBIC sensor platform. The BBIC is a small (2 cm square) hybrid Bio/Silico Photonic device consisting of engineered bioluminescent microbial cells interfaced directly with an optical application specific integrated circuit. Bioluminescence is directly proportional (over a given range) to the target analyte, based on lux gene transcriptional fusions in the bioreporter organisms. The critical path to successful implementation of a long term stable and robust sensor system based on the BBIC is to create an on-chip environment to stably maintain an inducible, physiologically controlled platform from which the microbial bioreporter can functionally operate. This platform can integrate into existing rudimentary CMOS electronic components already demonstrating sensor device functionality. To accomplish the goals of the project, the research will focus on upgrading the currently used sensor platform in order to manage a stable BBIC environment. The prototype package will include the incorporation of negative and positive control bioluminescent test strains to accommodate false positive and negative system responses. The sensor platform is intended to service various analyte detection BBICs in a plug and play format for generally universal environmental monitoring. The specific objective of this proposal is to significantly upgrade current biochip (BBIC) technology in a format conducive to NASA's requirement for reliable and adaptable, low power/low mass environmental monitoring and control systems. The planned research and development brings together in a collaborative effort the University of Tennessee with two NASA research centers for the institution of defined modifications to integrated circuit performance for increased sensitivity, improved reliability, and enhanced power management. With concurrent development of bioreporter constructs for specific, sensitive detection of emphasized Priority 1 pollutants, an established deployable microelectronic sensor technology will be created for system wide multiarray monitoring of crew habitats. These research efforts will yield networks of BBIC sensors capable of remote, unmanned profiling of the chemical environment
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The detection of chemicals using miniaturized autonomous biosensors can be applied in office buildings, homes, and other enclosed structures (submarines, for example) to continuously monitor air quality (i.e., mold growth or other toxic buildups associated with sick building syndromes) and for the detection of chemical warfare agents.
Task Progress:	Design and fabrication of two separate integrated circuit BBIC systems has been completed. Both systems have simple interfaces, requiring no external biasing networks. The first of the two systems is an op-amp integrator-based system fabricated in the TSMC 0.35-um process. The TSMC 0.35-um system provides an on-chip DAC for tuning of the system. The second system is a Schmitt trigger based system fabricated in the AMI 0.5-um process. Testing was performed on both systems to characterize leakage (dark) current performance. The systems showed comparable dark current measurements on the order of 50 fA (50 ? 10-15 A). The TSMC 0.35-um op-amp system was tested using the 5RL bio-reporter strain. Results followed predicted patterns of bio-reporter response. The AMI 0.5-um chip also includes an integrated temperature sensor that provides a serial digital output that can be used for calibration of the BBIC system output. The sensor is based on the same Schmitt trigger architecture used on the second BBIC system and has been characterized over a 100 C range from 50 C to 50 C. Development of a system for characterization of the photodiodes has been completed. Using this system, quantum efficiency for the TSMC 0.35-um system photodiode was measured over the visible light spectrum. A quantum efficiency of approximately 55% was measured for TSMC 0.35-um photodiode at the bio-reporter wavelength of 490 nm. Two bioreporters are being developed; one for the detection of dichloromethane and the other for formaldehyde. The gene for dichloromethane sensing (dcmR/dcmA) and the gene for formaldehyde sensing (adhI) have been linked to the lux operon in an E. coli host. These genetic constructs now need to be removed and placed into an appropriate host strain (Methylobacterium dichloromethanicum for dichloromethane and Rhodobacter sphaeroides for formaldehyde). The parameters for electroporating the gene fusions into these strains are currently being developed. Protocols for binding bacteria to borosilicate glass have been developed using antibodies and glutaraldehyde as the linkage agents. Briefly, the techniques involve cleaning borosilicate glass followed by silanizing the surface using aminoproplytrimethoxysilane. Then either the antibody or glutaraldehyde were allowed to adhere to the silanized borosilicate. Bacteria were allowed to adhere to the surface in a 2 hour incubation. After the incubation step, the luminescence produced was read on a Wallac 1450 Microbeta Plus. The bacterial test strain is an E. coli with a constituitive lux operon. Both methods showed a maximum luminescence reading of approximately 1.2 X 106 counts per second (cps) with a background of 3000 cps.
Bibliography Type:	Description: (Last Updated: 03/16/2017)
Abstracts for Journals and Proceedings	Islam, S.; Weathers, B.; Terry, S.; Zhang, M.; Blalock, B.; Caylor, S.; Ripp, S.; Sayler, G. "Genetically-engineered whole-cell bioreporters on integrated circuits for very low-level chemical sensing" European Solid-State Device Research Conference, Grenoble, France XXXX , Sep-2005
Abstracts for Journals and Proceedings	Ripp S, Garland J, Sayler G. "Bioluminescent bioreporter integrated circuit sensing of the chemical and biological spacecraft environment" Presented at the 13th International Symposium on Bioluminescence and Chemiluminescence, Yokohama, Japan, August 2-6, 2004 . Luminescence. May/June;19(3):170-1. , Jun-2004
Books/Book Chapters	Ripp, S.; Garland, J.; Blalock, B.; Islam, S.; Sayler, G. "Bioluminescent bioreporter integrated circuit sensing of the chemical and biological spacecraft environment" in "Bioluminescence and Chemiluminescence Progress and Perspectives" Ed. Tsuji, A.; Matsumoto, M.; Maeda, M.; Kricka, L.; Stanley, P. World Scientific, Hackensack, New Jersey, 433-436 , Jun-2005